SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: MLA0301
COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Data Interpretation **Duration: 1hr**

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

(5 Marks)

Scenario: A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what it indicates about the learning performance and policy improvement of the agent.

1. Identify the Relevant RL Concept

The main concepts are reward signal, episodic performance, policy improvement, learning progress, and convergence. In Reinforcement Learning, the reward received by an agent provides feedback about how well its actions perform in the environment.

2. Examine the Given Data / Observation

The reward increases gradually from 10 at the beginning of training to 45 after further episodes. This is a positive upward trend. The agent is therefore receiving increasingly better outcomes from its interaction with the environment.

3. Analyze the Relationship with RL Behavior

At the start, the agent has limited knowledge of the environment and may select actions that produce low rewards. Through repeated interaction, the agent updates its value estimates or policy. As useful actions become more strongly associated with higher expected returns, the agent becomes more likely to select them.

Figure 1. Illustrative upward reward trend during RL training.

4. Interpret Using RL Principles

An increasing reward generally indicates that learning is taking place. The agent is improving its estimate of action values or its policy, depending on the algorithm used. A higher reward means that the agent is achieving more desirable states or completing the task more effectively.

5. Compare the Outcomes

The difference between the initial reward of 10 and the final reward of 45 is 35 reward units. The final reward is 4.5 times the initial value. This substantial improvement suggests that the learned behavior is much better than the behavior observed during early training.

6. Justification

The gradual rather than sudden improvement is also meaningful. It suggests incremental learning through repeated experience. If the reward later becomes approximately stable around a high value, that would provide stronger evidence that the policy is approaching convergence.

7. Conclusion

The increase in reward from 10 to 45 indicates positive learning performance and policy improvement. The agent is learning to select actions that generate higher returns. Therefore, the reward trend provides evidence that the policy is becoming more effective, although reward increase alone cannot prove that the policy is globally optimal.

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance. (5 Marks)

Scenario: An epsilon-greedy agent is tested with ε = 0.1, 0.3 and 0.5. The cumulative reward is highest when ε = 0.1 and lowest when ε = 0.5. Justify the result using exploration vs exploitation and identify the better epsilon value.

1. Identify the Relevant RL Concept

The key concept is the exploration–exploitation trade-off. In an epsilon-greedy strategy, ε controls the probability of exploration. With probability ε, the agent explores a non-greedy action; with probability $1 - ε$, it exploits the currently best-known action.

2. Examine the Given Data / Observation

The observed ranking is: ε = 0.1 gives the highest cumulative reward, ε = 0.3 gives an intermediate result, and ε = 0.5 gives the lowest reward. Thus, increasing epsilon in this scenario reduces the cumulative return.

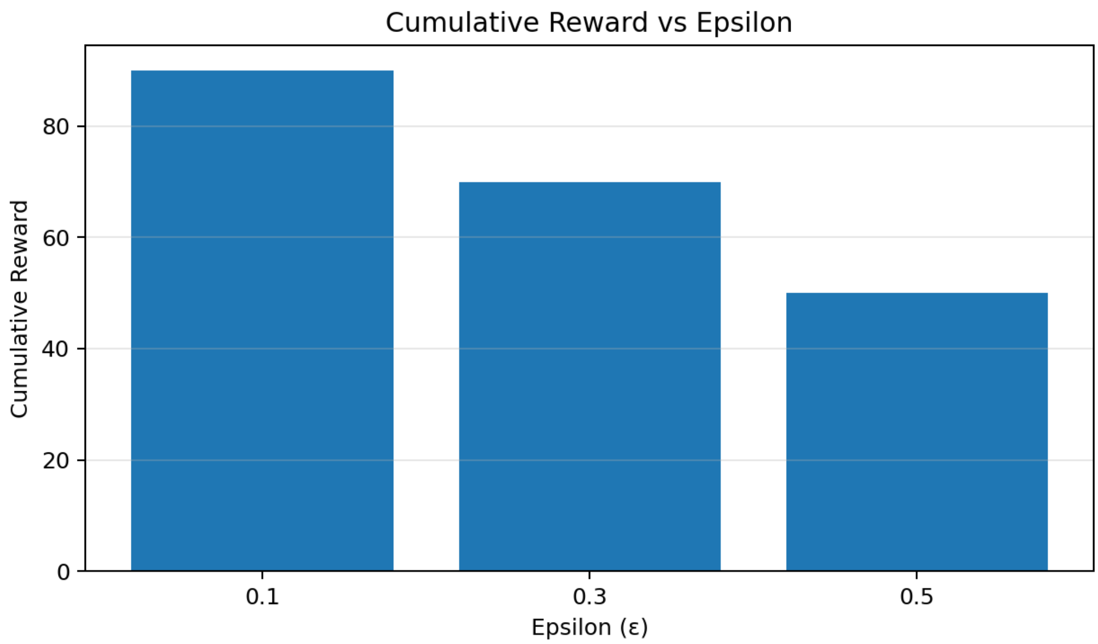

Figure 2. Illustrative comparison of cumulative reward for different epsilon values.

3. Analyze the Relationship with Agent Behavior

At ε = 0.1, the agent explores only occasionally and exploits the action that currently has the best estimated value most of the time. This is beneficial when the learned value estimates are already reasonably reliable. At ε = 0.5, exploration occurs very frequently, so the agent spends about half of its decisions trying alternatives instead of consistently selecting the best-known action.

4. Interpret Using RL Principles

A higher epsilon is not automatically better. Exploration is useful for discovering potentially better actions, especially early in learning. However, excessive exploration can lower immediate cumulative reward because the agent repeatedly selects actions that may be inferior. Conversely, too little exploration can cause the agent to miss better actions.

5. Compare the Outcomes

Among the tested values, ε = 0.1 achieves the highest cumulative reward, while ε = 0.5 achieves the lowest. Therefore, the tested environment favors greater exploitation after the agent has learned useful action preferences.

6. Justification

The result demonstrates a practical exploration–exploitation trade-off: ε = 0.1 provides enough exploration to avoid completely deterministic behavior while allowing the agent to exploit high-value actions most of the time. ε = 0.5 introduces substantially more random exploration, which lowers performance in this particular scenario.

7. Conclusion

The best-performing tested value is ε = 0.1 because it produces the highest cumulative reward. It offers a better balance between limited exploration and strong exploitation for the given environment. The best epsilon is environment- and training-dependent, so the result should not be generalized to every RL problem.

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results. (5 Marks)

Scenario: Q-Learning and SARSA are compared using average rewards over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference and determine which algorithm converges faster based on the observed results.

1. Identify the Relevant RL Concept

The relevant concepts are temporal-difference learning, on-policy learning, off-policy learning, Q-value updates, average reward, and convergence. Q-Learning is an off-policy TD control algorithm, whereas SARSA is an on-policy TD control algorithm.

2. Examine the Given Data / Observation

The stated observation is that Q-Learning has slightly higher average rewards at 100, 200 and 300 episodes. The difference is consistent across the three checkpoints.

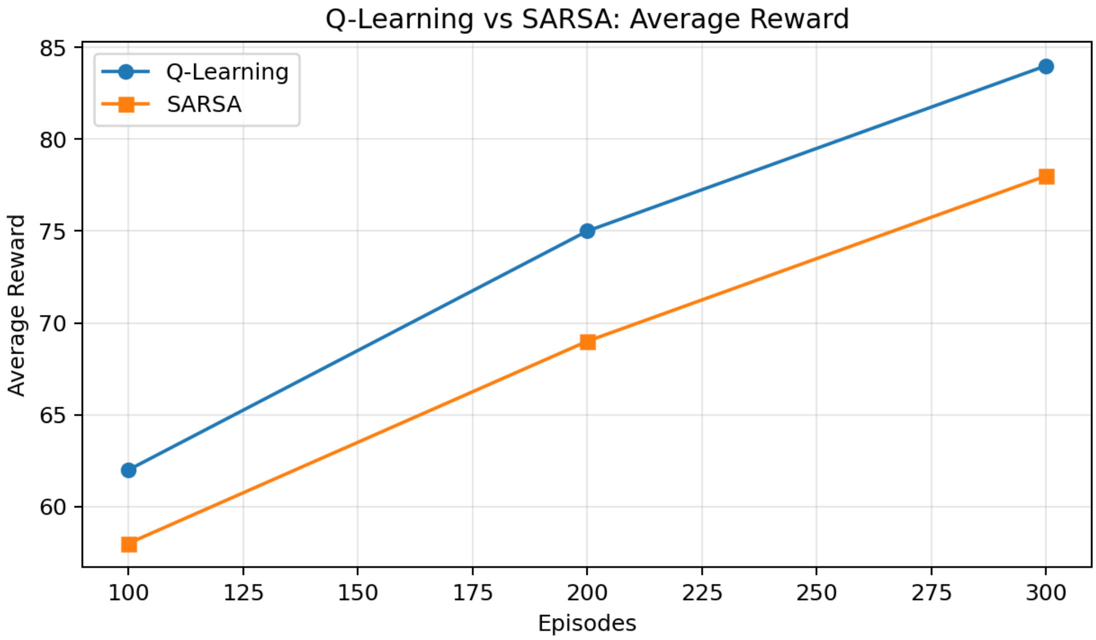

Figure 3. Illustrative average-reward comparison between Q-Learning and SARSA.

3. Analyze the Algorithm Behavior

Q-Learning updates its action value using the maximum estimated value of the next state, represented conceptually by max Q(s',a'). SARSA instead updates using the action that is actually selected under the current policy. Because Q-Learning learns toward the greedy target, it can sometimes learn a stronger greedy policy more quickly in a suitable environment.

4. Interpret Using RL Principles

The consistently higher reward of Q-Learning means that, for the given experiment, its learned policy is producing better returns at the observed checkpoints. SARSA can behave more conservatively because its update accounts for the action actually chosen, including exploratory actions.

5. Compare the Outcomes

At 100, 200 and 300 episodes, Q-Learning remains above SARSA. This consistent advantage suggests that Q-Learning is improving its policy at a faster rate in the reported experiment.

6. Determine Convergence

Based only on the observed results, Q-Learning appears to converge faster because it reaches higher reward levels at each checkpoint. Strictly speaking, faster convergence should be confirmed using a convergence criterion such as stabilization of Q-values or average reward, rather than a single comparison of reward values.

7. Conclusion

Q-Learning shows slightly better performance than SARSA in the given scenario and is the algorithm that appears to converge faster based on the observed reward trend. The result is consistent with Q-Learning's off-policy, greedy-target update, but actual convergence speed remains dependent on the environment, learning rate, exploration strategy, and initialization.

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. (5 Marks)

1. Identify the Relevant RL Concept

The key concept is the discount factor, γ. It determines how strongly future rewards are valued relative to immediate rewards. The return can be represented as $G_t = R_{(t+1)} + \gamma R_{(t+2)} + \gamma^2 R_{(t+3)} + \ldots$.

2. Examine the Given Data / Observation

The average reward increases as γ changes from 0.2 to 0.5 and then to 0.9. Therefore, the agent performs best among the tested values when it gives greater importance to future consequences.

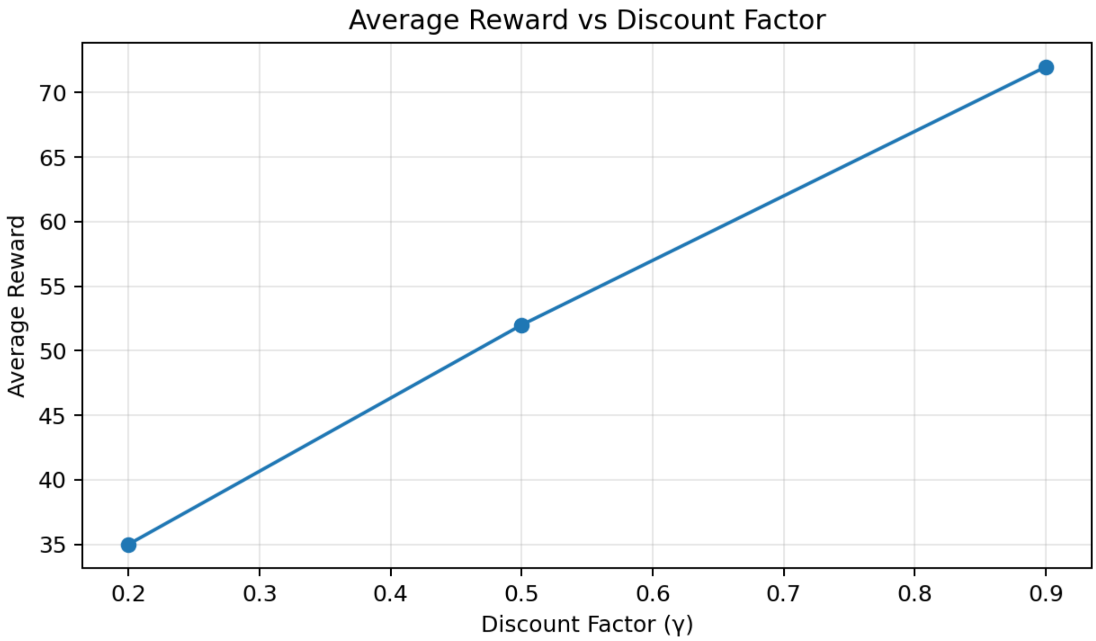

Figure 4. Illustrative relationship between discount factor and average reward.

3. Analyze the Relationship

With $\gamma = 0.2$, future rewards are strongly discounted, so the agent focuses mainly on immediate outcomes. With $\gamma = 0.5$, future outcomes have a moderate influence. With $\gamma = 0.9$, future rewards retain much more of their value, encouraging the agent to choose actions that may produce smaller immediate rewards but larger long-term returns.

4. Interpret Using RL Principles

A high discount factor is particularly useful in tasks where achieving the final objective requires a sequence of actions. The agent can learn that an action is valuable because it leads to a favorable future state, even if that action does not produce a large immediate reward.

5. Compare the Outcomes

The ordering of performance is $\gamma = 0.9 > \gamma = 0.5 > \gamma = 0.2$. This shows that, for the given environment, valuing future rewards more strongly improves the observed average return.

6. Justification

The discount factor should not simply be maximized in every RL problem. A very high γ can make learning more sensitive to long-term consequences and can sometimes slow or destabilize learning. However, among the three values provided, $\gamma = 0.9$ is most suitable because it gives the highest observed average reward.

7. Conclusion

The increase in reward with γ demonstrates that long-term planning is important in the given task. Therefore, $\gamma = 0.9$ provides the best learning performance among the tested values because it balances immediate and future rewards in favor of the long-term objective.

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. (5 Marks)

Scenario: The success rate of an RL agent increases from 45% at 50 episodes to 92% at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend.

1. Identify the Relevant RL Concept

The relevant concepts are success rate, policy improvement, training performance, generalization, and convergence toward an effective policy. Success rate measures how often the agent completes the desired task successfully.

2. Examine the Given Data / Observation

The success rate rises from 45% at 50 episodes to 92% at 200 episodes. This is an improvement of 47 percentage points. The agent therefore becomes successful in nearly all evaluated trials by the end of the reported training period.

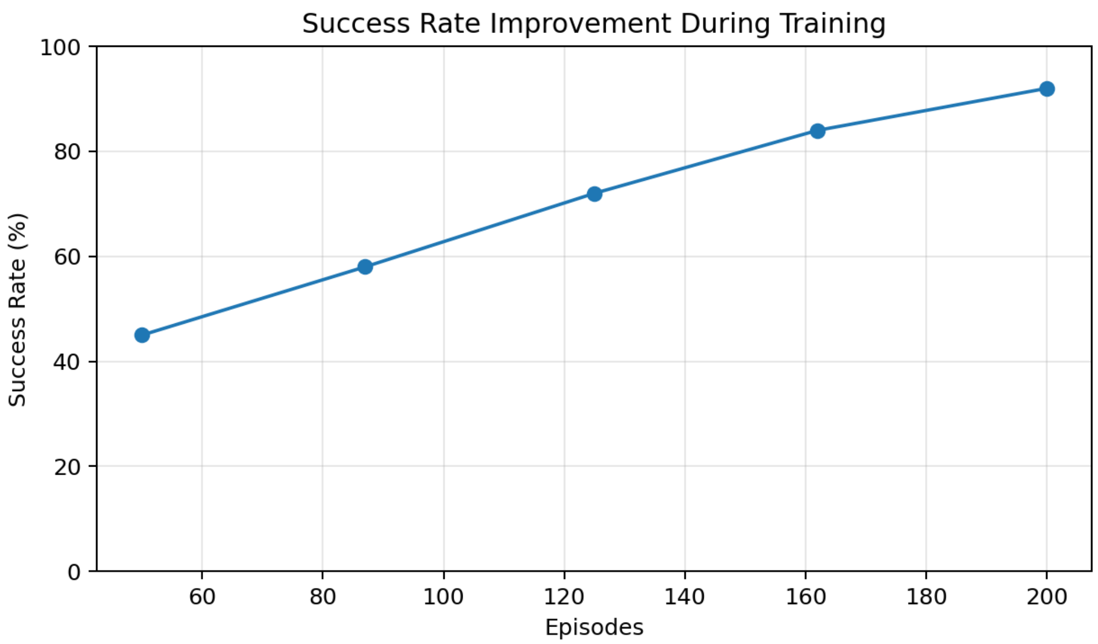

Figure 5. Illustrative increase in success rate as training episodes increase.

3. Analyze the Relationship with RL Behavior

During early training, the agent has limited experience and its policy may select ineffective actions. With more episodes, the agent receives additional reward feedback and updates its value estimates or policy. The rising success rate indicates that these updates are leading to increasingly effective decisions.

4. Interpret Using RL Principles

A 92% success rate is strong evidence of substantial policy improvement. It suggests that the agent has learned a policy that performs the task effectively in the evaluated environment. However, an observed success rate of 92% does not by itself prove that the policy is mathematically optimal.

5. Compare the Outcomes

At 50 episodes, fewer than half of the trials are successful. At 200 episodes, approximately nine out of ten trials are successful. The large and consistent improvement indicates that additional experience has significantly improved the agent's decision-making.

6. Justification

To claim an optimal policy, further evidence would normally be required, such as stable performance over additional episodes, comparison with the theoretical optimum or a strong baseline, and evaluation on independent episodes. If the 92% rate remains stable and is close to the maximum achievable performance, it would provide stronger evidence of convergence toward an optimal or near-optimal policy.

7. Conclusion

The increase from 45% to 92% clearly demonstrates strong learning and policy improvement. The agent has likely learned an effective policy, but the available data are insufficient to prove that it is fully optimal. A stable success rate near the maximum achievable level over further evaluation would strengthen the conclusion.

Overall Conclusions

The five selected scenarios demonstrate major Reinforcement Learning principles: reward improvement reflects policy learning; epsilon controls the exploration–exploitation trade-off; Q-Learning and SARSA differ in their update policies; the discount factor controls the importance of future rewards; and success rate is a practical indicator of task performance.

Question	Main RL Concept	Conclusion
1	Reward trend & policy improvement	Reward 10→45 indicates strong learning progress; high stable reward would support convergence.
2	Exploration vs exploitation	ε=0.1 performs best in the given scenario because it favors exploitation while retaining limited exploration.
3	Q-Learning vs SARSA	Q-Learning has higher observed rewards and appears to converge faster in this experiment.
4	Discount factor γ	γ=0.9 is best among tested values because it values future rewards and gives the highest average reward.
5	Success rate & policy learning	45%→92% indicates strong policy improvement, but does not alone prove global optimality.

Reinforcement Learning – Conceptual Background

Reinforcement Learning (RL) is a machine learning paradigm in which an agent learns how to make decisions by interacting with an environment. Unlike supervised learning, the agent is not given the correct action for every situation. Instead, it receives a reward or penalty after taking an action and gradually learns which actions are more useful. The objective is to learn a policy that maximizes the expected cumulative reward over time.

An RL system normally contains four important elements: the state, action, reward, and policy. The state represents the current situation of the environment. The action is a decision selected by the agent. The reward is numerical feedback received after an action. The policy is the strategy that maps states to actions. In many algorithms, a value function or action-value function is also learned to estimate how useful a state or state-action pair is.

The five scenarios selected in this assignment are useful because they represent common ways of evaluating RL learning. Reward curves show whether performance is improving. Epsilon experiments demonstrate the balance between exploration and exploitation. Comparisons between Q-Learning and SARSA show how different temporal-difference update rules affect learning. Discount-factor experiments show how future rewards influence decision making.

Finally, success-rate trends provide an intuitive measure of whether the learned policy is becoming effective.

When interpreting an RL experiment, it is important not to rely on one number alone. A high reward in one episode may occur by chance, particularly in a stochastic environment. Instead, average reward over several episodes, moving averages, success rate, episode length, and the stability of these measurements provide stronger evidence. A good analysis therefore considers both the direction of the trend and the consistency of the trend.

Important Reinforcement Learning Terminology

Several technical terms are useful for understanding the scenarios in this assignment.

State (s): The state describes the current condition of the environment from the agent's perspective. For example, in a maze, the agent's current position can be considered a state.

Action (a): An action is a decision available to the agent. In a maze, possible actions may include moving up, down, left, or right.

Reward (r): Reward is the feedback returned by the environment. Positive reward generally encourages useful behavior, while negative reward discourages undesirable behavior.

Policy (π): A policy specifies how an agent chooses actions. A deterministic policy selects one action for a state, while a stochastic policy assigns probabilities to multiple actions.

Return (G): Return is the cumulative reward considered by the agent, normally including discounted future rewards. It is different from a single immediate reward.

Value function V(s): The value function estimates the expected return from a state when following a particular policy.

A**ction-value function Q(s,a):** The Q-function estimates the expected return when taking action a in state s and then continuing according to a policy.

Exploration: Exploration means trying actions whose outcomes are uncertain in order to discover potentially better behavior.

Exploitation: Exploitation means selecting the action currently believed to have the highest value.

Learning rate (α): The learning rate controls how strongly newly obtained information changes an existing value estimate. A very large learning rate can make learning unstable, while a very small rate can make learning slow.

Discount factor (γ): The discount factor determines how strongly future rewards are considered. Values near zero emphasize immediate rewards, while values near one place greater importance on long-term returns.

Convergence: Convergence means that important learning quantities, such as value estimates or performance measures, become sufficiently stable. High reward alone does not automatically prove convergence; stability over repeated evaluations is important.

How Reward Curves Are Interpreted in RL

Reward is one of the most commonly used metrics for evaluating a reinforcement learning agent. During training, the agent interacts with the environment repeatedly. Each interaction produces rewards, and these rewards can be accumulated over an episode. If the average reward gradually increases, it usually indicates that the agent is discovering more useful actions.

A typical early-stage reward curve may contain large fluctuations. This is normal because the agent initially has little knowledge and may explore many actions. As learning progresses, value estimates become more informative and the policy becomes better adapted to the environment. A successful training process often shows an upward trend followed by a region where the reward fluctuates around a relatively high level.

However, an increasing reward curve must be interpreted carefully. Reward scale depends on the environment. For example, a reward increase from 10 to 45 is meaningful within the given experiment, but it does not allow direct comparison with another environment whose reward ranges from -100 to 100. The most important evidence is the relative improvement under the same environment and evaluation procedure.

The shape of the curve also matters. A smooth upward trend suggests gradual learning, whereas a highly unstable curve may indicate excessive exploration, an unsuitable learning rate, sparse rewards, or a difficult environment. If the reward becomes stable after improvement, it may indicate that the agent's policy has reached a region of good performance.

For the first question, the increase from 10 to 45 is therefore interpreted as evidence of positive learning. The agent is obtaining better outcomes as it gains experience. A stronger conclusion about convergence would require the reward to remain near the higher level for additional episodes.

Exploration and Exploitation in Greater Detail

The exploration–exploitation dilemma is one of the central issues in Reinforcement Learning. The agent must decide whether to use an action it already believes is good or try another action that might turn out to be better. If it always exploits, it may become trapped in a suboptimal policy. If it explores too much, it may fail to take advantage of knowledge it has already acquired.

Epsilon-greedy is a simple and widely used strategy. The parameter epsilon represents the probability of selecting an exploratory action. A small epsilon such as 0.1 means that approximately 10 percent of decisions are exploratory and the remaining decisions generally exploit the current best action. An epsilon of 0.5 produces substantially more exploration.

The best epsilon depends on the stage of learning and the environment. During early training, more exploration can be valuable because the agent has not yet discovered the useful actions. During later training, reducing exploration often improves performance because the agent can exploit what it has learned. This is why many practical systems use an epsilon-decay schedule, starting with a relatively high value and gradually reducing it.

In the given scenario, epsilon = 0.1 produces the highest cumulative reward. This suggests that the environment benefits from exploitation and that the current Q-value estimates are sufficiently useful. Epsilon = 0.5 produces the lowest reward because the agent spends more decisions exploring alternatives. This does not mean that exploration is bad; rather, it means that excessive exploration reduces cumulative reward under the particular test conditions.

A fair experiment should ideally compare the same number of episodes, random seeds, environment conditions, and evaluation procedure for every epsilon value. The reported cumulative reward should preferably be averaged across multiple runs because one training run can be affected by randomness.

Q-Learning and SARSA – Theoretical Comparison

Q-Learning and SARSA are both temporal-difference control algorithms, but they differ in the target used for updating action values. This distinction is important when interpreting their performance.

Q-Learning is an off-policy algorithm. Its update moves the current estimate toward a target based on the maximum estimated action value in the next state. In simplified form, the target contains max_a Q(s',a). This means that the update assumes the best next action will be taken, even if the current behavior policy sometimes explores other actions.

SARSA is an on-policy algorithm. Its name comes from the sequence State–Action–Reward–State–Action. The update uses the action that the current policy actually chooses in the next state. Consequently, exploratory behavior can directly influence the learned values.

Because of this difference, Q-Learning may learn an aggressive greedy policy in environments where the maximum next-state value is a good target. SARSA may learn a safer or more conservative policy when exploratory actions can lead to poor outcomes. Neither algorithm is universally superior. Their performance depends on the environment, reward structure, exploration method, learning rate, and other hyperparameters.

In the assignment scenario, Q-Learning consistently has slightly higher average rewards at 100, 200, and 300 episodes. This is evidence that Q-Learning is performing better at the specified checkpoints. Since it is already achieving higher rewards at earlier checkpoints, it can reasonably be described as appearing to converge faster in this experiment.

Nevertheless, convergence should be defined carefully. If Q-Learning has a higher reward at every checkpoint but is still increasing rapidly at episode 300, the evidence is not enough to say it has fully converged. A stronger experimental analysis would plot the moving average reward and continue training until both performance and Q-values stabilize.

Discount Factor and Long-Term Decision Making

The discount factor gamma is important whenever an agent must make decisions whose consequences occur several steps in the future. The return can be expressed as the immediate reward plus discounted future rewards. Therefore, gamma determines the relative importance of short-term and long-term consequences.

When gamma is close to zero, the agent behaves more like a short-sighted decision maker. It strongly prefers immediate rewards because future rewards lose their value quickly. This can be suitable for tasks where immediate outcomes are most important.

When gamma is closer to one, future rewards retain more value. The agent becomes more willing to take an action with a small immediate reward if that action leads to a more valuable future state. This is particularly useful in navigation, games, planning, and other sequential decision problems.

The given experiment uses gamma values of 0.2, 0.5, and 0.9, and reports that average reward increases as gamma increases. The result indicates that long-term consequences are important in the environment. The agent benefits from planning beyond the next immediate reward.

The value gamma = 0.9 is therefore the best among the tested settings. It gives the highest observed average reward and provides a strong emphasis on future outcomes. However, gamma should not be selected solely by the principle that higher is always better. Very high discounting of the immediate reward can sometimes increase the effective planning horizon and make learning more difficult. Hyperparameter selection should therefore be based on experimental evaluation.

In a practical RL experiment, gamma should be tested together with learning rate and exploration settings. Changing several parameters at once can make it difficult to identify which parameter caused a performance difference. Controlled experiments are therefore important.

Success Rate as a Policy Evaluation Metric

Success rate is a particularly intuitive metric because it directly measures whether the agent completes the desired task. In a navigation task, success could mean reaching the destination. In CartPole, success may relate to maintaining the pole for a required duration. In a game, it may mean winning an episode.

The scenario states that success rate rises from 45 percent at 50 episodes to 92 percent at 200 episodes. The increase of 47 percentage points is substantial. It means that the agent has moved from failing in more than half of the evaluated trials to succeeding in nearly nine out of ten.

This improvement is consistent with policy learning. The agent has accumulated additional experience, updated its action values, and become more effective at choosing actions. If the success rate continues to remain around 92 percent across additional evaluation episodes, that would indicate stable performance.

However, optimality is a stronger claim than effectiveness. An optimal policy is one that achieves the best possible expected return under the defined objective and environment. A 92 percent success rate may be near-optimal in one environment, but it could still be below an achievable 100 percent success rate in another. Therefore, an analyst should compare the result with a theoretical optimum, expert baseline, or strong reference policy when available.

Another important consideration is evaluation bias. Training success rate should not be the only measure because the agent may have benefited from exploration during training. A separate evaluation phase with exploration reduced or disabled can provide a clearer estimate of the learned policy's quality.

Thus, the 45-to-92 percent trend is strong evidence of policy improvement. It supports the conclusion that the agent has learned an effective strategy, while additional evaluation is required before declaring the policy optimal.

Best Practices for Interpreting RL Experiments

A strong Reinforcement Learning analysis should distinguish between learning progress, performance, and convergence. Learning progress refers to whether the agent is improving through experience. Performance refers to how well the current policy performs. Convergence refers to whether important quantities have become stable.

First, repeated trials should be considered because RL contains randomness from initial states, environment transitions, and exploration. A single run may produce an unusually high or low reward. Reporting the mean and variation across multiple runs is more reliable.

Second, reward should be examined together with other metrics. For example, an agent might obtain high reward by taking a long route if the reward structure permits it. Episode length, success rate, and constraint violations can provide additional information.

Third, moving averages can reduce visual noise in reward curves. A moving average does not change the underlying data; it makes the general trend easier to observe. This is especially useful in stochastic environments.

Fourth, hyperparameters should be tested systematically. Epsilon, learning rate, discount factor, batch size, and network architecture can all affect performance. A fair comparison changes one important factor at a time whenever possible.

Finally, the evaluation policy should be separated from the training policy when appropriate. During training, exploration is useful. During evaluation, excessive exploration can make a good policy appear worse than it actually is. Clear evaluation procedures therefore make conclusions more trustworthy.

Code of Conduct:

I Malli Chandana(192472250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Malli chandana

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand